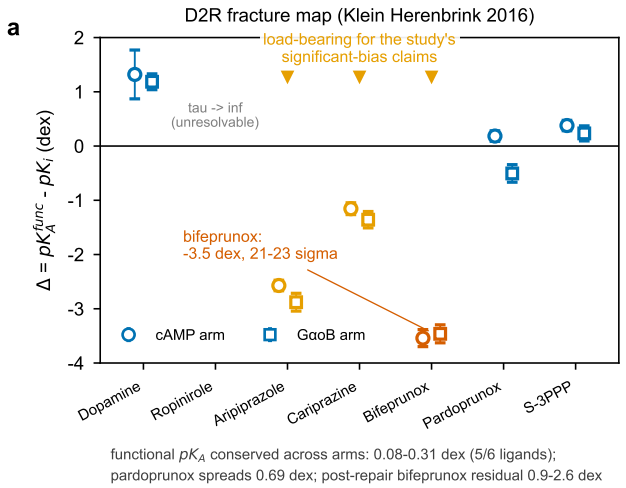
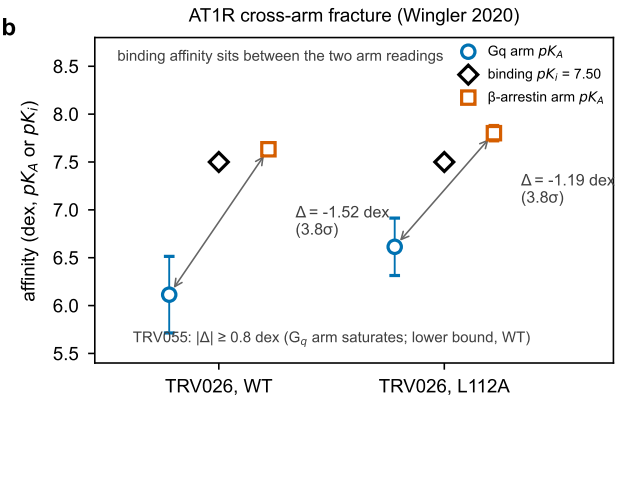
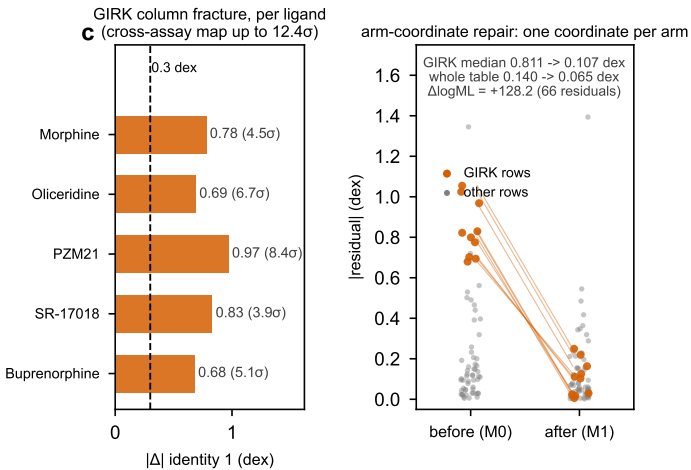
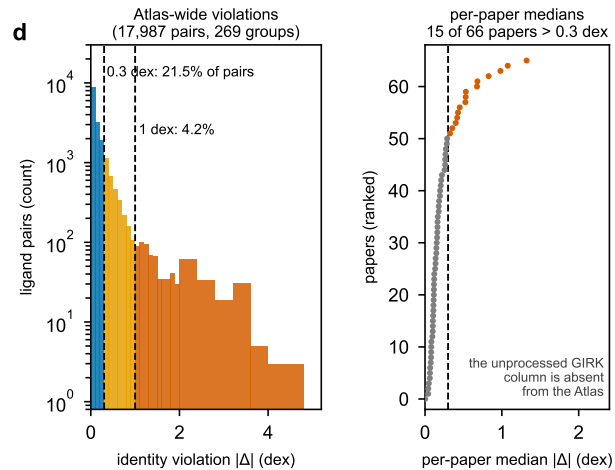

a
D2R fracture map (Klein Herenbrink 2016)
load-bearing for the study's
significant-bias claims
tau -> inf
(unresolvable)
bifeprunox:
-3.5 dex, 21-23 sigma
cAMP arm
GαoB arm
Δ = pK_A^func - pK_i (dex)
Dopamine
Ropinirole
Aripiprazole
Cariprazine
Bifeprunox
Pardoprunox
S-3PPP
functional pK_A conserved across arms: 0.08-0.31 dex (5/6 ligands);
pardoprunox spreads 0.69 dex; post-repair bifeprunox residual 0.9-2.6 dex


b
AT1R cross-arm fracture (Wingler 2020)
binding affinity sits between the two arm readings
Gq arm pK_A
binding pK_i = 7.50
β-arrestin arm pK_A
affinity (dex, pK_A or pK_i)
Δ = -1.52 dex
(3.8σ)
Δ = -1.19 dex
(3.8σ)
TRV055: |Δ| ≥ 0.8 dex (G_q arm saturates; lower bound, WT)
TRV026, WT
TRV026, L112A


c
GIRK column fracture, per ligand
(cross-assay map up to 12.4σ)
0.3 dex
Morphine
0.78 (4.5σ)
Oliceridine
0.69 (6.7σ)
PZM21
0.97 (8.4σ)
SR-17018
0.83 (3.9σ)
Buprenorphine
0.68 (5.1σ)
|Δ| identity 1 (dex)
arm-coordinate repair: one coordinate per arm
GIRK median 0.811 -> 0.107 dex
whole table 0.140 -> 0.065 dex
ΔlogML = +128.2 (66 residuals)
GIRK rows
other rows
|residual| (dex)
before (M0)
after (M1)


d
Atlas-wide violations
(17,987 pairs, 269 groups)
0.3 dex: 21.5% of pairs
1 dex: 4.2%
ligand pairs (count)
identity violation |Δ| (dex)
per-paper medians
15 of 66 papers > 0.3 dex
papers (ranked)
the unprocessed GIRK
column is absent
from the Atlas
per-paper median |Δ| (dex)